StarCraft AI: ProtoBot

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Senior Design

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Background

What is StarCraft: Brood War?

StarCraft: Brood War (1998) is a real-time strategy game developed by Blizzard Entertainment, where players gather resources, expand their influence across the map, and control units to defeat an opponent. Unlike turn-based games, everything happens in real time, requiring players to make quick decisions, manage multiple tasks, and adapt to incomplete information due to limited visibility of the opponent.

Why does AI in StarCraft matter?

StarCraft presents a challenging environment for AI because it combines real-time decision making, resource optimization, and strategic planning under uncertainty. Games like StarCraft also provide a way to be able to train agents without having to make complex machinery and only requires hardware to run online games to train. Artificial intelligence has already transformed how humans approach games like Chess, Go, and many more. Real-time games like *StarCraft: Brood War* represent the next level of complexity AI should strive solve to both exceed and teach human players new strategies, improving other AI fields along during the process.

Key Requirements

To make a competitive competition bot and provide the team guidelines for what our bot should achieve. The team constrained ProtoBot to adhere to well known StarCraft AI tournaments.

Sample of the AIIDE rules:

- Games will be a full 1v1 game of StarCraft 1.16.1 Brood War, fog of war enabled.
- Bot must be compiled with BWAPI and be able to run on Windows 10.
- 1 frame exceeding 10 seconds, 10 frames exceeding 1 second, or 320 frames exceeding 55ms will result in a loss for the bot.
- Games will have a frame limit of 86400 frames, approximately one hour of gameplay.
- A bot crashing or pausing the game will result in an immediate loss for it.
- Bots will be disqualified if they use perfect information or exploit any bugs.

While rules between competitions can vary, the AIIDE ruleset is used as a baseline for many StarCraft AI competitions and provides a good framework for what ProtoBot should achieve computationally.

Architectural Design

To make the creation of ProtoBot be as seamless as possible, the team utilized STARTcraft as a skeleton to structure ProtoBot. STARTcraft is an open-source repository created by Dr. Churchill at Memorial University of Newfoundland and serves as an accessible blank project for those interested in developing AI for *StarCraft: Brood War*. STARTcraft provides all the necessary BWAPI and StarCraft files needed to start developing an agent, allowing the team to test early iterations of our bot against StarCraft's native AI to develop basic behavior.

ProtoBot's architecture has a central system with subsystems that ensures logical data flow. The central system, **ProtoBot Commander**, handles these multiple subsystems, a.k.a. managers, that are each responsible for a particular task in the game.

- **Combat Manager**: Manipulates large groups of units to attack enemies and defend key positions.
- **Scouting Manager**: Controls individual combat units for scouting possible enemy locations and harassing/killing important targets
- **Economy Manager**: Assigns worker units to resources and optimizes their farming routes to maximize resource gains.
- **Information Manager**: Scans for enemy units, buildings, and bases to store their positions, counts, and current states for use in other managers
- **Build Manager** : Queues building placements and verifies unit production to strengthen ProtoBot's army.
- **Strategy Manager** : Analyzes current game state to issue strategic commands to other managers.

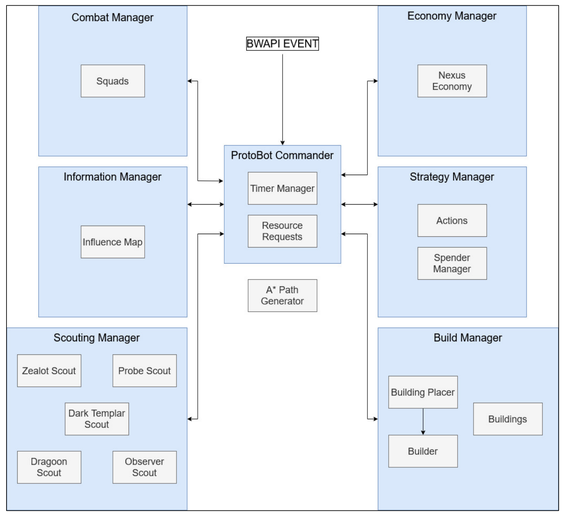

Figure 1: System Architecture of ProtoBot

Implementation Details and Test Plan

For StarCraft games to be played with code, a software known as the Brood War API (BWAPI) was developed to allow C++ and many other languages to interface with StarCraft. This allows code to command actions and get limited details about the events of a StarCraft game.

ProtoBot listens for 11 BWAPI events to command each module to execute specific actions. Through the combination of these events and each manager, ProtoBot can execute actions in a StarCraft game and adapt to changes in a StarCraft environment.

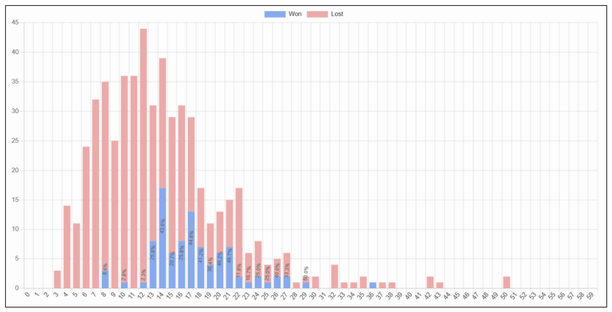

Figure 2: ProtoBot v1.0 BASIL Per Game Winrate

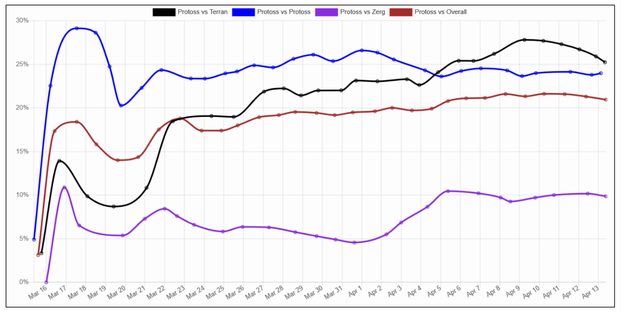

Figure 3: ProtoBot v1.0 BASIL Wins Per Race

For testing during senior design 1, the team would run games against the native AI created by Blizzard Entertainment. Providing quick feedback about the basic level of ProtoBot's gameplay and provided an environment where the team could address crashing due to miss handling of unit references, edge cases, StarCraft bugs, etc.

Once ProtoBot was in a non crashing state, during the time between sprints 6 and 7, ProtoBot was uploaded to an online ladder called BASIL. BASIL is a 24/7 hosted ladder where over a million games have been played by over 100 submitted bots. The ladder provides bot authors a way to test changes before big competitions and provides a way for new bot authors to test their agent against others. Uploading to the ladder also had the added benefit for authors to provide feedback about ProtoBot through the community Discord.

Conclusions and Future Work

ProtoBot demonstrates a competitive, student-made *StarCraft: Brood War* bot that can win matches without crashes or interrupts. ProtoBot serves to help future senior design teams understand the scale of projects involving game-playing agents in real-time strategy games. As well as highlighting the applications of games as environments for AI research.

Some Future Improvements Involve:

- Fine-tuning small-scale combat (target prioritization, formation variety, etc.)
- Tailoring unit production to the weaknesses of the enemy's current units
- More adaptable placement of buildings
- Optimization of ProtoBot's long computation times of specific algorithms.

References

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